

Open and Closed Neighborhood Balance Numbers of Graphs

Micaela Roth

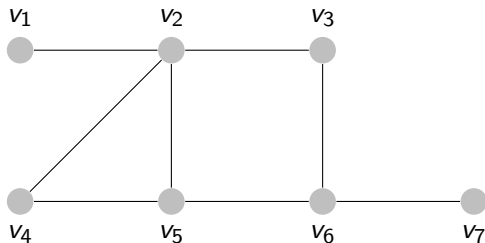
Thesis Advisor: Prof. Ann Trenk

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Motivation: Workplace

Suppose each employee in a workplace has several collaborators. We want to teach the employees two different skills so that there is an optimal balance of skills among each person's collaborators.



► People = vertices; collaborators joined by lines; skills = colors.

Question: is it possible for each person's circle of collaborators to have an even balance of skills?

Outline

1. Motivations: Workplace
2. Definitions: Graph, 2-Coloring, Red and Blue Degree
3. Previous Work: NBC and CNBC Graphs
4. The balance number for graph families
5. The edge balance number
6. Conclusion + Open Questions

Definitions

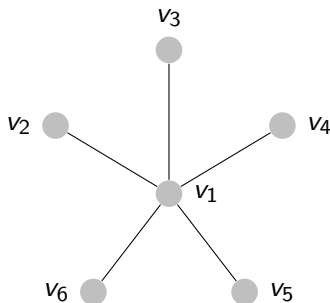
Defn: Graph

We define a *graph* G as a collection $V(G)$ of vertices and $E(G)$ of edges, which are unordered pairs of elements from V . Two vertices are *adjacent* if they are connected by an edge.

Example:

$$V(G) = \{v_1, v_2, v_3, v_4, v_5, v_6\}$$

$$E(G) = \{v_1 v_2, v_1 v_3, v_1 v_4, v_1 v_5, v_1 v_6\}$$



Definitions

Defn: 2-coloring

A 2-coloring of a graph G is an assignment of a color in the set $\{\text{red}, \text{blue}\}$, to each vertex in $V(G)$.

Defn: Open and closed neighborhoods

The open neighborhood of $v \in V(G)$ is:

$$N(v) = \{u : uv \in E(G)\}$$

and its closed neighborhood is $N[v] = N(v) \cup v$.

Defn: Red and blue degrees

Given a 2-coloring of G and some $v \in V(G)$, the open and closed “red and blue degrees of v ” are the number of red or blue vertices in their open and closed neighborhoods, respectively.

- Notation example: the closed red degree of v is $\text{deg}_R[v]$.

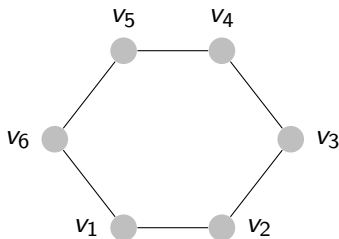
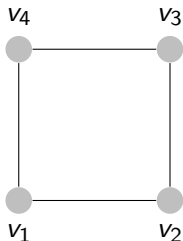
Definitions

Defn: Open and closed neighborhood balanced coloring

Given a graph G , an open/closed neighborhood balanced coloring is a 2-coloring of G such that the open/closed neighborhoods of each vertex have the same number of red and blue vertices.

Defn: NBC and CNBC graphs

An NBC graph is a graph that has an open neighborhood balanced coloring, and a CNBC graph is a graph that has a closed neighborhood balanced coloring.



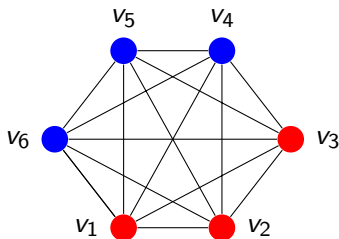
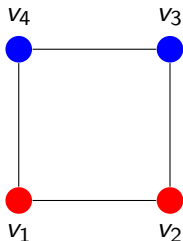
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Previous Work

Timeline:

- ▶ Idea first introduced by Freyburg and Marr in 2024 article: “Neighborhood balanced colorings of graphs”
- ▶ Closed balance case: “Closed Neighborhood Balanced Coloring of Graphs” by Collins et al. in 2025.
- ▶ Several papers generalizing to 3 colors or k colors.
- ▶ Recent work on generalizing: “Color 2-switches and neighborhood λ -balanced graphs with k colors” by Collins et al. in 2026.

Limitations of NBC and CNBC graphs

- ▶ To be NBC, every vertex degree must be even. To be CNBC, every vertex degree must be odd.
- ▶ We might want to be able to optimize this balance of colors for any graph.

The balance number of a graph

A 2-coloring of a graph G is open- λ -balanced at vertex $v \in V(G)$ if:

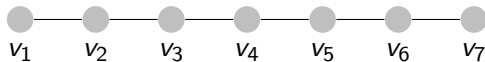
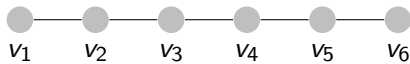
$$|\deg_R(v) - \deg_B(v)| \leq \lambda$$

A 2-coloring of G is open λ -balanced if it is open λ -balanced at all $v \in V(G)$.

- The balance number $b(G)$ is the minimum λ for which G is λ -balanced.

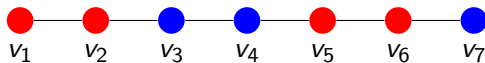
The closed balance number $b[G]$ is defined analogously using closed neighborhoods and degrees.

Example: Paths



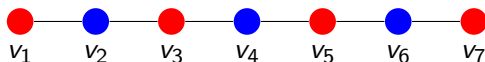
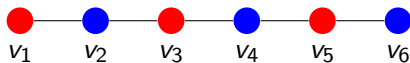
$$b(P_n) = ?$$

Example: Open balance number for paths



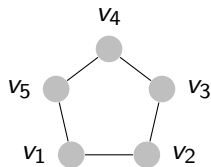
$$b(P_n) = 1$$

Example: Closed balance number for paths

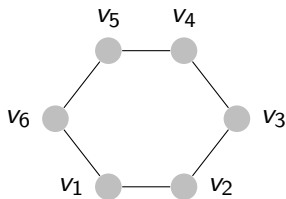


$$b[P_n] = 1$$

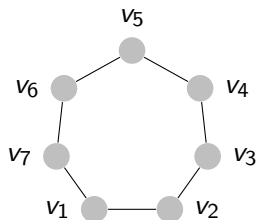
Example: Open balance number for cycles



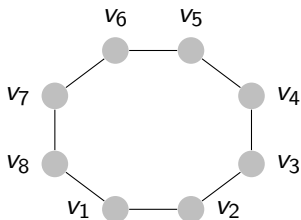
C_5



C_6



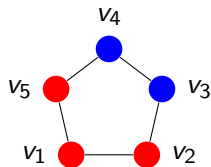
C_7



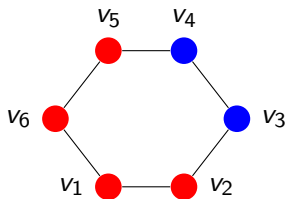
C_8

$$b(C_n) = ?$$

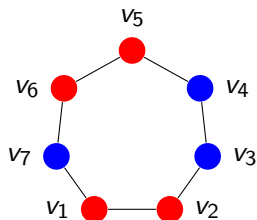
Example: Open balance number for cycles



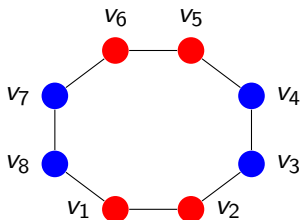
C_5



C_6



C_7



C_8

$b(C_n) = 0$ if $n \equiv 0 \pmod{4}$, otherwise $b(C_n) = 2$.

Balance number of a hypercube

Defn: Hypercube

The *hypercube* Q_n has vertices $\{(x_1, \dots, x_n) : x_i \in \{0, 1\}\}$, and has an edge between two vertices iff they differ in exactly one coordinate.

- Visualization: Q_n is a cube in n dimensions.

Hypercubes

If n is even, then $b(Q_n) = 0$, and if n is odd, then $b(Q_n) = 1$.

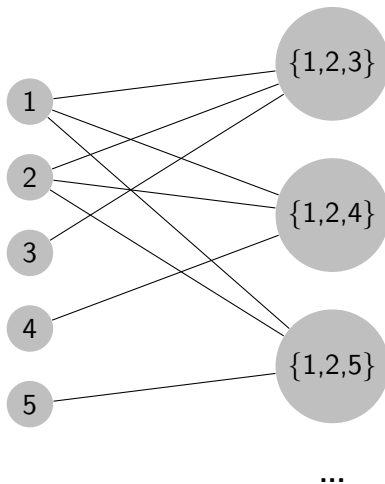
Proof sketch:

Color Q_n where a vertex $\mathbf{x} = (x_1, x_2, \dots, x_n)$ is red if $\sum x_{2i}$ is odd and blue if $\sum x_{2i}$ is even.

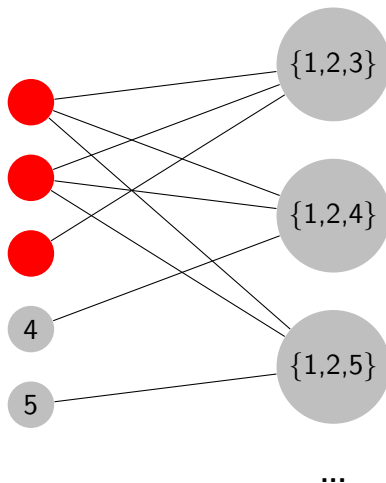
- Neighbors are off by one coordinate, half of the time this is an even coordinate.
- Example of neighbors: $(0, 1, 1, 0, 1, 0)$ and $(0, 1, 0, 0, 1, 0)$

Constructing a graph with arbitrarily large balance number

Example: $\lambda = 3$



Constructing a graph with arbitrarily large balance number



To generalize, for balance number $b(G) \geq \lambda$ we want $2\lambda - 1$ vertices on the left and $\binom{2\lambda-1}{\lambda}$ subset vertices on the right.

The edge balance number

Training employees skills in pairs $\rightarrow$ color edges using red and blue.

- Define edge neighborhoods, degrees, edge λ -balanced, and edge balance number analogously.

Defn: Edge red and blue degrees

Given an edge 2-coloring of a graph G , we define:

$$\deg'_R(v) = \text{num of red edges incident to } v$$

$$\deg'_B(v) = \text{num of blue edges incident to } v$$

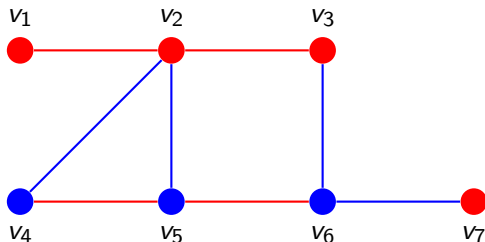
Defn: Edge balance number

The *edge balance number* of G is the minimum value across edge colorings of the largest difference for some $v \in V(G)$:

$$|\deg'_R(v) - \deg'_B(v)|$$

Using a λ -balanced vertex coloring to generate a λ -balanced edge coloring ($\lambda = 1$)

Color an edge red if its two endpoints have the same color, and blue otherwise.



- ▶ As a consequence of this construction, we have $b'(G) \leq b(G)$ for any graph G .
- ▶ This inequality does *not* hold in the other direction.

Graphs with $b'(G) = 0$

By the inequality $b'(G) \leq b(G)$, if a graph G has $b(G) = 0$, then $b'(G) = 0$ as well. We also state the following sufficient condition:

Thm: Edge balance number of a bipartite graph

If G is a bipartite graph and each vertex has even degree, then $b'(G) = 0$.

Proof:

Use Euler cycles.

Necessary condition: if $b'(G) = 0$, then each vertex has even degree.

Additional Results

- ▶ Open and closed balance numbers or bounds for trees, wheels, complete graphs, complete bipartite graphs, unit interval graphs, threshold graphs

Some Open Questions

- Find $b[T]$ for all trees T .
- We found the edge balance numbers of a few different graph families. This work can be extended by finding the edge balance numbers of other graph families.
- We established some necessary and sufficient conditions for graphs with $b'(G) = 0$. Can we come up with alternative or stronger conditions?
- An alternate version of balancing an edge 2-coloring is to balance the set of edges that are incident to each *edge*. However, an open question is to similarly classify the edge balance numbers of different graph families using this version of the definition.

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Thank you for listening! Questions?